

Is the surface forcing through sea ice leads transferred to the Arctic Ocean interior?

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Abstract

The Arctic sea ice, in particular the ice pack, acts as an insulator between the atmosphere and the ocean below. Leads, common in Arctic sea ice, facilitate exchanges of heat, salt, moisture, and gases between the ocean and the atmosphere. Local observations have captured heat fluxes through some leads at least an order of magnitude larger than those outside of the leads, leading to the speculation that air-sea exchanges through leads may contribute significantly to the Arctic Ocean surface buoyancy forcing. However, due to the lack of observations and high-resolution models capable of resolving leads, an estimate of the impact of leads on the ocean buoyancy forcing and water mass transformation has been elusive. Here, we quantify the seasonal magnitude and impact on the ocean surface caused by the leads using SEDNA, a subkilometer pan-Arctic hindcast (1/60°; NEMO model). Leads account for 22% of the sea-ice cover extent, and within them, there is approximately 25% of the total surface water mass transformation. The buoyancy flux and water mass transformation in leads are similar to those underneath the surrounding ice-covered oceans and the percentage of water mass transformation through leads matches their area coverage. Thus, the present estimate indicates that leads have a small contribution to Arctic Ocean dynamics, contrary to previous hypotheses.

Keywords: Arctic, Leads, Buoyancy flux

047 The Arctic sea ice regulates Earth’s climate by acting as a natural insulator between
048 the atmosphere and the ocean (Wettlaufer et al, 1997; Untersteiner, 1961). A ubiqui-
049 tous feature of Arctic sea ice is the formation and persistence of leads. Leads occur
050 across the polar oceans, both in the marginal ice zone (MIZ; 15 – 80% sea ice concen-
051 tration) and the ice pack (> 80% sea ice concentration). Within the MIZ, leads are
052 primarily formed by the advection of sea ice, while in the ice pack, leads form through
053 deformation of the sea ice and are commonly referred to in literature as “linear kine-
054 matic features” (LKF). The primary forcing of the sea ice advection and deformation
055 is the wind (Linow and Dierking, 2017; Hutter et al, 2018), with a smaller contribu-
056 tion from ocean currents (Willmes et al, 2023). Leads result in openings of the sea
057 ice cover (Rampal et al, 2016; Hutchings et al, 2005; Richter-Menge et al, 2002) with
058 spatial scales of meters to kilometers in width and a few kilometers up to hundreds
059 of kilometers in length, and temporal scales ranging from a few hours up to a few
060 days (Linow and Dierking, 2017; Wernecke and Kaleschke, 2015; Tschudi et al, 1998).
061 Thus, they can locally impact the ocean surface forcing in the ice-covered polar oceans
062 (Lüpkes et al, 2008).

063 Once leads are formed, atmospheric forcing in conjunction with localized upwelling
064 or downwelling occurring in the surface layer can result in important heat fluxes at
065 the ocean surface ($\mathcal{O} \sim 100W/m^2$; Bourgault et al, 2020; Marcq and Weiss, 2012;
066 McPhee et al, 2005; Maykut, 1986), instigating localized melting or freezing of sea ice
067 (von Albedyll et al, 2022). For example, Boutin et al (2023) estimated that 35% of the
068 total sea ice growth during winter occurs within the leads in the ice pack. The Arctic
069 ice-covered ocean is a β -ocean, i.e. the ocean stratification is mainly controlled by
070 salinity, which in turn is largely determined by ice-ocean interactions (sea-ice growth
071 and melt). Thus, leads experiencing an increase in buoyancy forcing due to brine
072 rejection can induce convection (Smith and Morison, 1998), weaken the mixed layer
073 stratification, and generate fronts, mixed layer turbulence, and eddies (Reiser et al,
074 2020; Smith et al, 2002). Meanwhile, leads experiencing melting will increase the mixed
075 layer stratification and stabilize the ocean surface layer. These changes in the buoyancy
076 flux translate into local transformation of the surface water masses that could be
077 critical for the functioning of the Arctic Ocean circulation (Lenn et al, 2022; Pemberton
078 et al, 2015; Smith et al, 1990). Previous studies have shown that instances of leads
079 impact locally the ocean surface and the properties of the Arctic Ocean mixed layer.
080 Nonetheless, the integrated contribution of leads to the large-scale buoyancy forcing
081 at the Arctic Ocean surface has not been quantified yet.

082 Here, we present the first estimate of the contribution of the buoyancy forcing
083 and water mass transformation within leads in the Arctic Ocean, using the last year
084 of a seven year long pan-Arctic hindcast run at a subkilometer resolution (SEDNA,
085 Talandier and Lique, 2023; see Methods for more details). Our main objectives are
086 to: (1) assess the impact of leads in the surface buoyancy forcing and (2) contrast the
087 surface water mass transformation within the leads and those of the ice cover excluding
088 leads.

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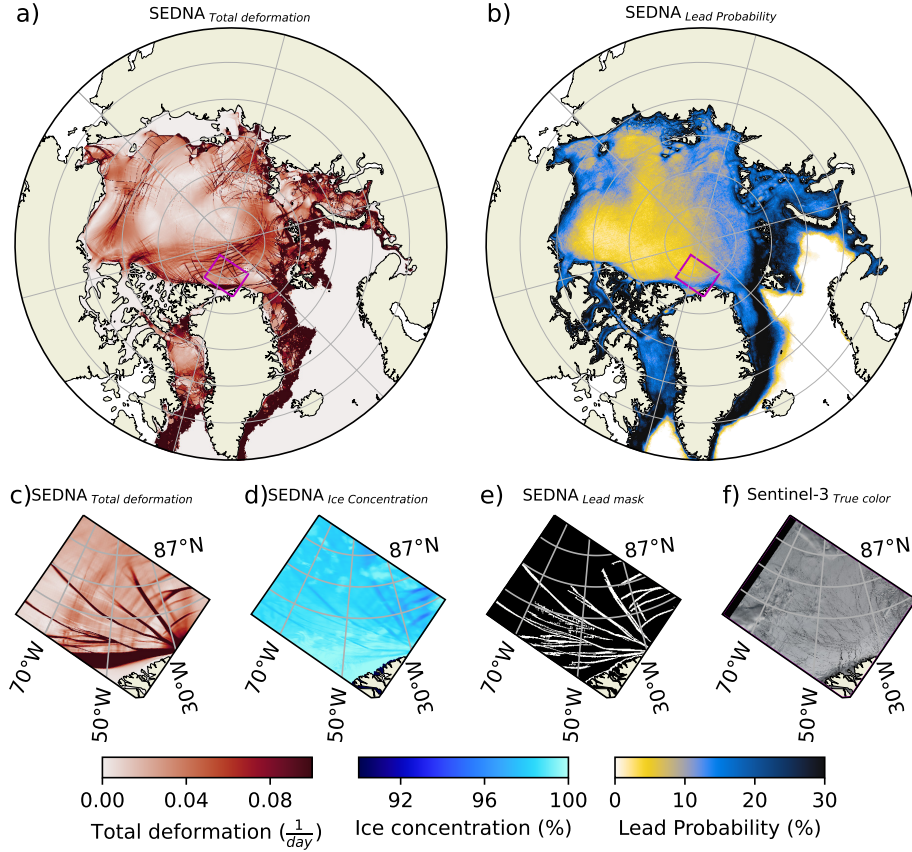


Fig. 1 a) Snapshot of total deformation for the 21st of April 2014 from SEDNA. b) Mean lead probability for daily snapshots of leads identified between 2011 and 2015 from SEDNA. Zoom of the magenta box of panel a for the c) total deformation to identify leads, d) ice concentration, and f) identified mask of leads on the 21st of April 2014 in SEDNA. e) True color image from Sentinel 3 on the 12th of April 2022 for the magenta area in panel a.

1 Buoyancy forcing in leads

Leads can be easily visually identified when examining high resolution satellite observations (Figure 1f), yet it remains challenging to capture them in state-of-the-art models that often lack the resolution and/or the sea ice dynamics to simulate these features (Bouchat et al, 2022; Wang et al, 2016). Here we analyze daily outputs from the SEDNA ocean-sea ice model, which has an average Arctic resolution of $\sim 800m$, sufficient to capture the observed lead width distribution ($> 1km$; Wernecke and Kaleschke, 2015). We start our analysis by detecting the leads in the model sea ice fields, using the detection algorithm of Hutter et al (2019), that identifies leads based on the sea ice deformation field (see methods). Figure 1a and c shows a snapshot of the sea ice deformation for the 21st of April 2014, from which we retrieve a mask of the leads (Figure 1e). Within the leads, the sea ice concentration can remain higher

139 than 90%; this is a consequence of the continuous nature of the sea ice model and the
140 daily averaged output that smooths the presence of leads in the fields.

141 The SEDNA climatology of the lead probability, i.e. the likelihood of leads occur-
142 ring for any given day, is shown in figure 1b. The average over the Arctic amounts
143 to $\sim 19\%$, comparable to previous observations (Wernecke and Kaleschke, 2015;
144 Willmes et al, 2023). Although observations use different identification methods com-
145 pared to the one implemented in SEDNA (Hutter et al, 2019), there is resemblance
146 between SEDNA and observations (Fig. S1), with similar lead probability over the
147 shelves, along bathymetric features, and in regions with large kinetic energy (Fig.
148 S1b). SEDNA reproduces the overall spatial patterns of lead probability observed in
149 the Arctic, except for the Beaufort Sea. Regardless, the similarities between the obser-
150 vations and the model lead probability give us confidence to quantify for the first time
151 the impact of leads on the surface buoyancy of the Arctic Ocean.

152 Figure 2a and b show two snapshots of the buoyancy flux at the ocean surface. A
153 negative buoyancy flux results from a loss of heat to the atmosphere sea ice refreezing
154 and brine rejection, while a positive buoyancy flux is associated with sea ice melting,
155 freshening, and warming of the Arctic surface. Inside the ice-covered region, the fluxes
156 are spatially homogeneous, except for the marginal ice zone and within the leads,
157 where the buoyancy fluxes are larger. The average buoyancy flux across the ice covered
158 Arctic (15% - 100% ice concentration) within the leads is $-2.2 \times 10^{-8} m^2/s^3$ and
159 $7.2 \times 10^{-8} m^2/s^3$ for each date, compared to the $-2.4 \times 10^{-8} m^2/s^3$ and $5.7 \times 10^{-8} m^2/s^3$
160 excluding the leads. In other words, on the 1st of January (winter), the total fluxes
161 within the leads and those outside of the leads are comparable. However, on the 25th
162 of June (summer), the average fluxes in the leads can be up to 30% larger than those
163 over the ice cover region excluding leads.

164 Examining a transect in the Canadian Basin on the 1st of January 2014, we detect
165 two leads (vertical blue bars) with a significant decrease in sea ice thickness of $\sim 25cm$,
166 suggesting an opening of the sea ice cover (Fig.2 c). This transect exhibits small heat
167 flux at the ocean surface and within the identified leads (Fig.2 e) because the ocean
168 surface is at freezing point and the heat exchanged with the atmosphere is used to
169 form sea ice rather than cooling the ocean. The freshwater flux (Fig.2g) exhibits two
170 prominent positive peaks located within the identified leads. Both peaks are twice the
171 magnitude of the fluxes outside the leads, highlighting enhanced sea ice growth and
172 brine rejection within the leads. The total buoyancy flux is negative for this snapshot,
173 with the largest buoyancy loss occurring within the leads (Fig.2 i). On the 25th of
174 June (Fig.2d), we detect several leads, but only some show a decrease of the sea ice
175 thickness of more than $1m$ (Fig.2 h). These specific leads also experience an increase in
176 heat fluxes, reaching up to $\sim 25W/m^2$ (Fig.2f) and a local decrease of the freshwater
177 flux, indicating a freshening of the surface due to sea ice melt. This change is captured
178 by the net positive buoyancy flux, decreasing the density at the surface (Fig.2j). Of all
179 the identified leads, some exhibit significant fluxes at the ocean surface, while others
180 have little to no imprint in the thickness and fluxes. This may be explained by the
181 timing of the sea ice break up, as subsequent snapshots show an increase of fluxes at
182 the location of these identified leads (not shown). Alternatively, previous studies have
183 shown that leads generated through shear have a limited buoyancy exchange with the
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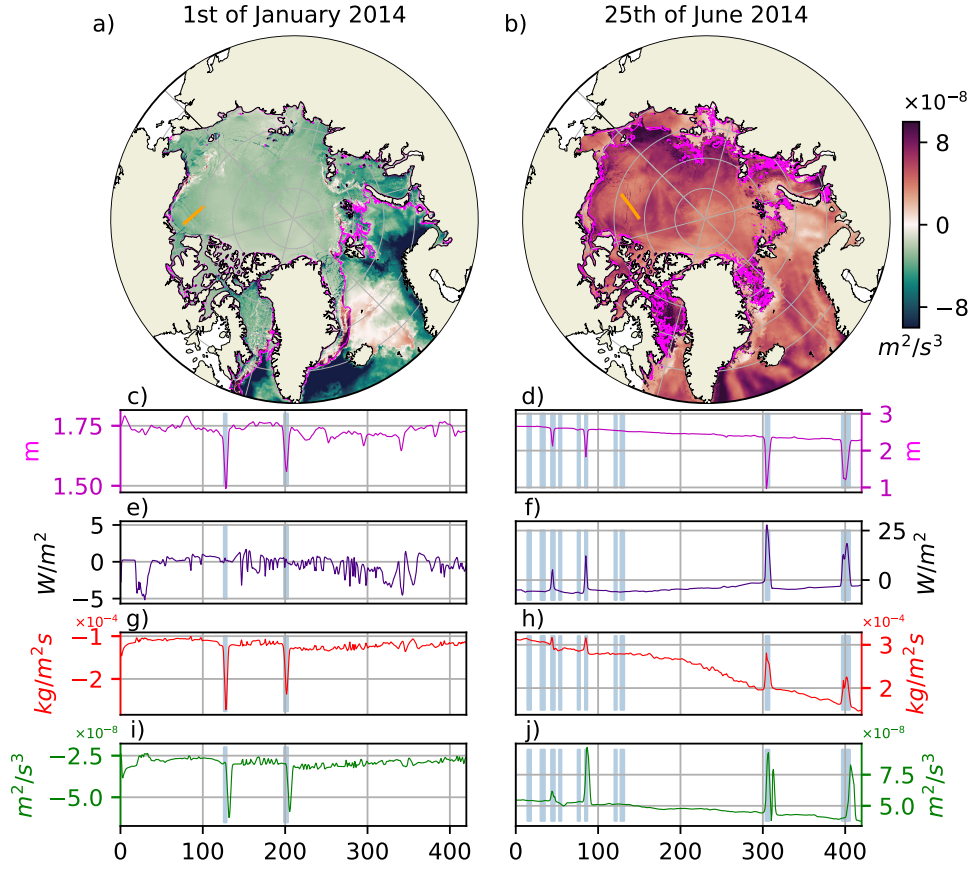


Fig. 2 Maps of buoyancy flux and transects for the 1st of January and 25th of June 2014. a) Map of buoyancy flux on the 1st of January 2014. Transect on the 1st of January 2014 of c) ice thickness, e) downward heat flux, g) freshwater flux (positive upward), and i) buoyancy flux (heat flux minus freshwater flux). b) Map of buoyancy flux on the 25th of June 2014. Transect on the 25th of June 2014 of d) ice thickness, f) downward heat flux, h) freshwater flux (positive upward), and j) buoyancy flux. Note that positive buoyancy fluxes correspond to buoyancy loss, while negative buoyancy fluxes correspond to buoyancy gain by the ocean. The transect location and orientation changes between the winter and summer dates. Magenta contour in panels a and b corresponds to the 15% concentration contour of sea ice. Vertical blue bars show the mask of leads in the transect.

atmosphere, but may still enhance fluxes in the ocean due to isopycnal upwelling or downwelling forced by the atmosphere-ocean stress within the leads (Bourgault et al, 2020).

Similar to observations, we identify fluxes that are larger within the leads than outside of them for these summer and winter days. In the next section, we explore the spatially averaged contribution of leads to the surface buoyancy fluxes and the integrated influence on the water mass transformation across the Arctic Basin during 2014.

2 Integrated contribution of leads to the Arctic buoyancy

The following analysis is conducted separately for four distinct masks representing regions with leads and regions excluding leads for the MIZ and the ice pack. On average over 2014, these masks account for 6% for the MIZ, 72% for the ice pack, 6% for the leads in the MIZ, and 16% for the leads in the ice pack. Thus the part covered by leads is $\sim 22\%$ of the total sea ice covered Arctic, consistent with previous model assessments (Wang et al, 2016). The extent of each mask exhibits a seasonal cycle (Fig. 3a). From winter (December, January, and February) to summer (June, July, and August), the ice pack extent decreases from approximately 80% to 60%, while the extent of the MIZ increases from 2% to approximately 13%. On one hand, the percentage of the ice pack leads extent peaks in winter at 16% and decreases to a minimum extent of 13% in summer. On the other hand, the extent of the MIZ leads covers 2% in winter and around 13% in summer, approximately the same as the MIZ extent.

The contribution of the total buoyancy flux occurring within the leads and ice covered regions is quantified by weighting the buoyancy flux of each of the sea ice cover masks by the total ice covered area (Fig. 3b; note that the size of each mask evolves in time). The mean buoyancy flux under the ice pack in winter (December-February) and summer (June-August) is $-1.8 \times 10^{-8} m^2/s^3$ and $2.2 \times 10^{-8} m^2/s^3$, respectively. The buoyancy flux underneath the ice pack changes signs during the seasonal cycle, suggesting a shift from ice formation in winter to ice melting in summer. Note that in the ice pack, the haline component of the buoyancy flux determines the buoyancy flux, while the heat contribution is negligible (See Fig.S2). Meanwhile, the mean buoyancy flux in the MIZ is negligible in winter and increases to $0.9 \times 10^{-8} m^2/s^3$ in summer. The MIZ has predominantly a positive buoyancy flux suggesting preferential melting throughout the year (except for autumn), and the haline component also determines the buoyancy flux (See Fig.S2). In both, the MIZ leads and the ice pack leads, the buoyancy flux follows the same seasonality of the fluxes outside of the leads. Winter values are $-0.3 \times 10^{-8} m^2/s^3$ and $-0.01 \times 10^{-8} m^2/s^3$, while summer values reach $0.6 \times 10^{-8} m^2/s^3$ and $0.9 \times 10^{-8} m^2/s^3$ for the contribution of the leads in the ice pack and the MIZ, respectively. Overall, leads explain between 20% to 45% of the ice covered buoyancy fluxes (red dashed line in Fig 3b), more than the area extent of all leads that ranges from 18% in winter to 26% in summer. Therefore, fluxes are larger within the leads relative to their area extent.

The role and impact leads have on the surface water masses of the Arctic can be quantified by using the surface water mass transformation framework (WMT; see methods section; Walin, 1982). This approach provides information on the transport of water through the surface density contours due to surface buoyancy forcing that results in the formation of lighter or denser water at the surface. Thus, the WMT relates the surface buoyancy forcing and the surface density field to the properties of the ocean interior. Figure 4a shows the averaged WMT during 2014 for the total ice covered Arctic, the ice pack, the MIZ, and the parts of these regions covered by leads. The definitions of the water masses depicted in figure 4a follow roughly those

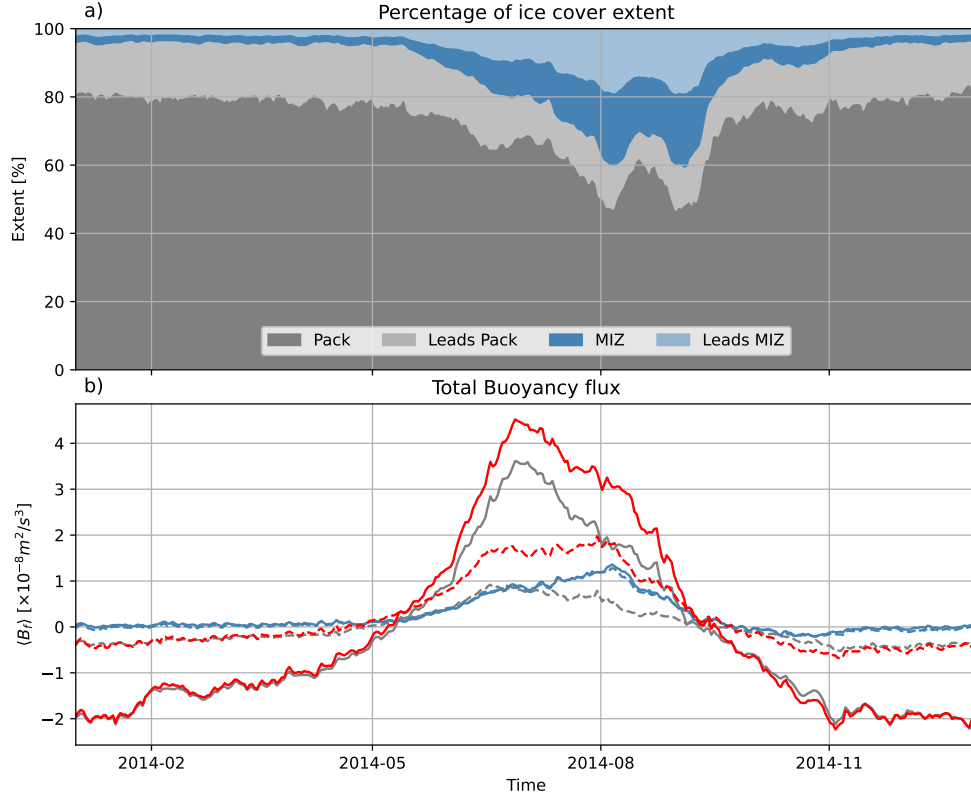


Fig. 3 Time series during 2014 of a) the percentage of ice cover extent and b) the total buoyancy fluxes for the total ice covered, the ice pack, the MIZ, and the leads in the ice pack and the MIZ. The MIZ is defined between ice concentration of 15-80%, the ice pack is defined where ice concentration > 80%, and the total ice covered is defined as the sum of the MIZ and ice pack.

proposed by Pemberton et al 2015 and Lansard et al 2012. The yearly mean WMT in the ice pack is characterized by a broad region of strong positive transformation (losing buoyancy) corresponding approximately to the halocline polar surface water (PSW_H ; $24.4kg/m^3 < \rho < 27.4kg/m^3$) reaching a maximum of $\sim 1.8Sv$, consistent with previous estimates of the surface Arctic WMT (Pemberton et al, 2015). The yearly mean WMT in the ice pack leads also features a positive peak within the PSW_H density class reaching up to $0.4Sv$. On average during 2014, the WMT in the ice pack leads is $\sim 20\%$ of the WMT in the ice pack (without leads). The mean MIZ WMT is mostly negative within the density range of $20kg/m^3 < \rho < 27.5kg/m^3$ (gaining buoyancy), reaching a minimum of $-0.4Sv$ around $\rho = 27kg/m^3$. The MIZ leads WMT is also negative within the same range of densities, with a minimum of $-0.4Sv$. Thus, the MIZ leads WMT is comparable to those in the MIZ excluding leads. Hovmöller diagrams of the daily WMT estimates over 2014 for the four masks are depicted in figure 4. The mean WMT underneath the ice pack varies from 1.1 Sv

in winter to -1.0 Sv in summer, while the WMT in the leads in the ice pack is 0.2 Sv in winter and -0.26 Sv in summer. In other words, the WMT in the ice pack leads ranges from 20 to 25% of the WMT occurring in the pack (Fig. 4d). It is interesting to note that the patterns across the density classes of the WMT for the leads and the surrounding regions excluding leads are similar (figures 4b and c). This suggests that there is no specific water mass that is transformed preferentially with the leads. Rather, our results suggest that the water masses transformed within the leads and outside of them have similar properties. The WMT in MIZ leads and those in the MIZ WMT have comparable magnitudes over the year (Fig. 4 g). Similarly to the leads in the ice pack, there is no evidence that leads in the MIZ could transform preferentially specific water masses compared to the remaining MIZ. Adding the contribution of the WMT in all the leads in the Arctic (those in the MIZ and the ice pack), we estimate $\sim 25\%$ of the WMT happens in leads, and this is comparable to the surface covered by these leads in the Arctic ($\sim 22\%$).

3 Conclusions

The present study assesses the impact of leads on the ocean surface by diagnosing the buoyancy fluxes using a very high-resolution hindcast (SEDNA). SEDNA captures the intermittency of the leads and the lead properties. Leads cover $\sim 22\%$ of the sea-ice covered Arctic, and in the ice pack leads can contribute more buoyancy flux compared to the fluxes outside the leads. Despite these larger buoyancy fluxes, leads only account for approximately $\sim 25\%$ of the total surface water mass transformation in the ice covered Arctic, which is roughly equivalent to their surface coverage. Thus, contrary to previous hypotheses based largely on local and intermittent observations (e.g. Hutter et al (2019); McPhee et al (2005)), our findings suggest that leads have a smaller contribution to the transformation of surface water masses in the Arctic Ocean. While their presence can induce localized changes in the density of surface waters due to brine rejection and ice melting, the transformation of water masses in the leads has the same sign and magnitude as those underneath the surrounding ice-covered regions. Thus, no evidence was found that leads impact in a distinct way the Arctic Ocean water masses in SEDNA compared to outside the leads in the ice covered ocean.

Here we solely focus on the surface forced WMT, but we acknowledge that leads could further impact properties at the surface and interior of the ocean by enhancing mixing within leads, inducing upwelling of isopycnals through Ekman pumping (McPhee et al, 2005), modifying entrainment in the mixed layer within the leads, providing a source of potential energy, and generating mixed layer instabilities due to isopycnal tilting in the vicinity of leads. The use of a continuous sea ice model that results in leads with high ice thickness and concentrations likely underestimates the exchanges between the ocean and the atmosphere. Therefore, future studies using floe-resolving models that allow more “realistic leads” with direct exchanges between the ocean and the atmosphere may result in larger fluxes through the leads increasing their potential impact in the water mass transformation and ocean mixed layer.

Leads have an important local effect in the atmosphere and the sea-ice fluxes (Marcq and Weiss, 2012; Lüpkes et al, 2008; Wettlaufer et al, 1997; Smith et al,

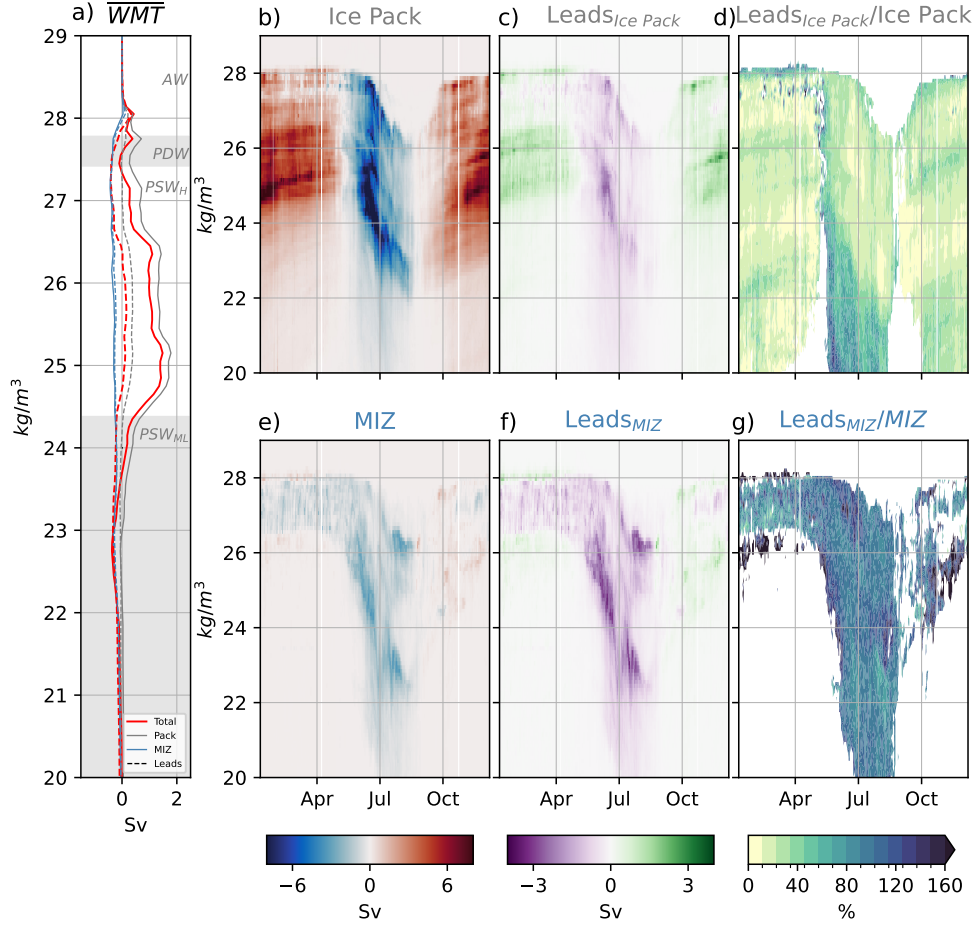


Fig. 4 Mean and daily Water Mass Transformation (WMT) over 2014. a) Mean water mass transformation over 2014 for the total ice covered, ice pack, MIZ, and leads in each region. Hovmöller diagram of the water mass transformation (Sv) for b) the ice pack, c) the ice pack leads, and d) the ratio of water mass transformation between the leads and the ice pack region. Panels e), f), and g) follow the same structure as panels b), c), and d), but for the MIZ, MIZ leads, and the ratio between them. Shaded gray and white areas in panel a) represent approximately the water masses of the Arctic Ocean according to [Pemberton et al 2015](#) and [Lansard et al 2012](#): the mixed layer polar surface water (PSW_{ML} ; $\rho < 24.4 \text{ kg/m}^3$), the halocline polar surface water (PSW_H ; $24.4 \text{ kg/m}^3 < \rho < 27.4 \text{ kg/m}^3$), the polar deep water (PDW ; $27.4 \text{ kg/m}^3 < \rho < 27.8 \text{ kg/m}^3$), and the Atlantic water (AW ; $27.8 \text{ kg/m}^3 < \rho$).

1990), but a limited imprint on the ocean surface and interior properties. Our results suggest that the presence of leads is not the first order mechanism controlling the water mass transformation forced at the surface, thus leads have a minor impact on the stratification and surface layer dynamics of the Arctic Ocean. Therefore, resolving and/or parameterizing leads in climate models are likely to only marginally improve

415 the misrepresentation of the stratification and water masses found in Arctic climate
416 models (Wang et al, 2024; Ilicak et al, 2016). Finally, the probability of leads occurring
417 in the ice pack has been projected to increase as the Arctic sea ice transitions towards
418 thinner, younger, and more mobile sea ice (Boutin et al, 2023; Intergovernmental Panel
419 on Climate Change (IPCC), 2022). Our analysis is only performed over one year and
420 the future impact of more abundant leads is not represented in our simulation. Thus,
421 dedicated studies are required to better estimate the surface buoyancy forcing within
422 leads in the context of an evolving Arctic Ocean.

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424 4 Methods

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426 Model description

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428 Here we employ the NEMO numerical platform (release 4.0.5; Madec et al, 2022) to
429 build a 1/60° pan-Arctic ocean-sea ice configuration named SEDNA. The ocean core
430 (NEMO-OPA) solves the primitive equations using finite differences on an Arakawa C-
431 grid mesh, and it is coupled to the SI3 sea-ice component (NEMO-SI3; NEMO Sea Ice
432 Working Group, 2022). The pan-Arctic domain covers the Bering Strait on the Pacific
433 side and extends southward to 56°N in the Subpolar North Atlantic and to the Baltic
434 Sea entrance. The ocean model incorporates a linear free surface, utilizes a 3rd order
435 flux form scheme for momentum advection, and employs a 4th order Flux Corrected
436 Transport (FCT) for tracer advection. Additionally, the sea-ice representation uses an
437 Elasto-Visco-Plastic (EVP) rheology with a 5-categories sea ice thickness distribution
438 and a landfast ice parameterization to better simulate sea-ice behavior above shelves.

439 The NCAR bulk formula from Large and Yeager (2009) is used to calculate air-sea
440 fluxes based on the hourly ERA5 atmospheric data for near-surface variables (Hersbach
441 et al, 2020). The three lateral boundaries are constrained with daily GLORYS12V1 Re-
442 analysis data (Jean-Michel et al, 2021). Monthly freshwater fluxes from river discharges
443 are combined with Greenland land ice melt data (Hu et al, 2019). The simulation
444 extends over seven years (2009 to 2015) and starts from rest, utilizing initial conditions
445 based on the World Ocean Atlas 2009 temperature/salinity and mean January 2009
446 ice state from PIOMAS re-analysis (Locarnini et al, 2010; Zhang and Rothrock, 2003).
447 A weak restoring toward the World Ocean Atlas climatological sea surface salinity is
448 applied, but only over the ice-free regions. Preliminary assessment against available
449 observations and re-analyses data shows that the simulation reasonably captures key
450 ocean and sea-ice characteristics.

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Identification of leads

The leads from SEDNA are identified by using the algorithm proposed by [Hutter et al \(2019\)](#). This method uses the total deformation rate defined as:

$$T_d = \left[\underbrace{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2}_{Divergence} + \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2}_{Shear} \right]^{1/2}, \quad (1)$$

where u and v are the sea ice velocities. The deformation rate varies spatially depending on the background deformation and the sea ice properties ([Reiser et al, 2020](#)), where the largest deformation rates occur along the boundaries of ice floes where the leads are found. Therefore, to identify the leads, the local maxima is identified using a difference of Gaussian filter (DoG filter) of the deformation rate field. After this, a binary map is created with the mask of identified lead. Note that after this step, [Hutter et al \(2019\)](#) reduce the leads width to a line to then track the leads over time. Since our study focuses on the ocean-atmosphere interactions, we only use the identification algorithm to produce a mask of leads during 2014 for the Arctic, thus we retain the original lead width computed from the deformation rate. Finally, the lead mask is then used to quantify and contrast the impact of leads compared to the ice covered Arctic, the ice-pack (ice concentration > 80%), and the MIZ (0 - 80% ice concentration).

Buoyancy flux and water mass transformation

Heat fluxes (shortwave radiation, longwave radiation, latent heat, and sensible heat) and freshwater fluxes (evaporation, precipitation, runoff, and sea ice melt/growth) at the ocean surface result in a buoyancy forcing capable of changing the density of the ocean surface. Here, the buoyancy flux is computed as:

$$B_f = \underbrace{\frac{g\alpha Q_{net}}{\rho_0 C_p}}_{Net\ surface\ heat\ flux} - \underbrace{\frac{g\beta F_{net} SSS}{\rho_0}}_{Net\ surface\ freshwater\ flux}, \quad (2)$$

where Q_{net} the net heat flux at the sea surface, α the thermal expansion coefficient, SSS the sea surface salinity, F_{net} the net freshwater flux that includes evaporation, precipitation, runoff, and freezing minus melting flux, β the haline contraction coefficient, $g = 9.81m/s^2$, and the ocean density $\rho_0 = 1026kg/m^3$. Note that using this sign convention, positive values in the buoyancy forcing decrease the surface density making the ocean surface more buoyant (stable) and negative values increase the surface density. The area-weighted buoyancy ($\langle \rangle$) fluxes are computed following:

$$\langle B_f(t) \rangle = \frac{\sum_x \sum_y (B_f(t, x, y) * Area(x, y) * Mask_R(t, x, y))}{\sum_x \sum_y * Area(x, y) * Mask(t, x, y)}, \quad (3)$$

where the $Area$ is the area of the grid cells, and the $Mask_R$ can be the mask of MIZ, the Pack, the MIZ leads, the ice pack leads, and the ice cover that varies in space and

time. Note that the divisor *Mask* corresponds to the full ice covered area (including leads).

Nurser et al (1999) and Walin (1982) proposed the water mass transformation framework that quantifies the rate at which a given water mass is made lighter (negative transformation rate) or denser (positive) due to surface fluxes. The surface water mass transformation is computed as:

$$\Omega(\sigma_k) = -\frac{1}{\sigma_{k+1} - \sigma_k} \iint_A B_f dA, \quad (4)$$

where σ is potential density and the indices k represents the density bin number. The computation was performed with 172 density bins of $0.1\text{kg}/\text{m}^3$ within the density range of $15.15\text{kg}/\text{m}^3$ - $32.25\text{kg}/\text{m}^3$. The transformation rate Ω is spatially decomposed into different sea-ice masks such as the ice covered, the pack ice, the MIZ, and the leads in each of these regions.

Open Source. The SEDNA model configuration are described and publicly available at <https://zenodo.org/records/7656924>. All analyses and figures in this manuscript are reproducible via Jupyter notebooks and instructions can be found in the GitHub repository *Leads_Arctic_ocean* (https://github.com/josuemtzmo/Leads_Arctic_ocean).

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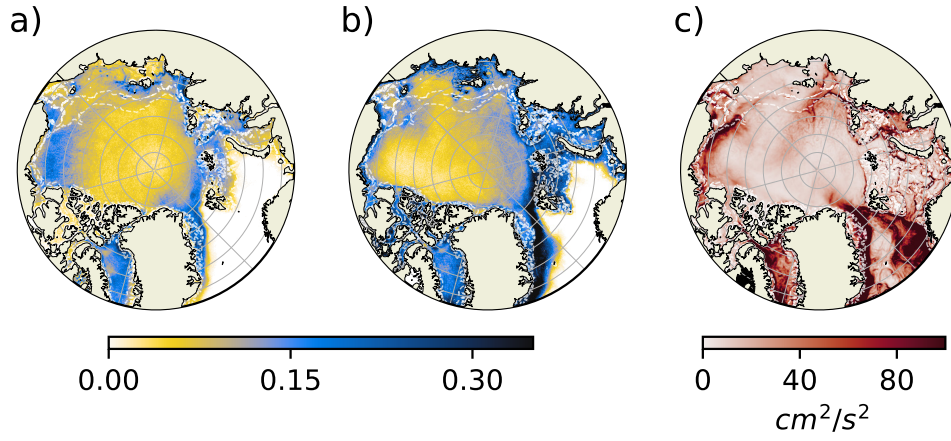


Fig. S1 Comparison of winter lead frequency between a) Thermal-Infrared Satellite Imagery (Willmes et al, 2023) and b) SEDNA between 2011 and 2015. Panel c) shows the SEDNA mean kinetic energy between 2011 and 2015.

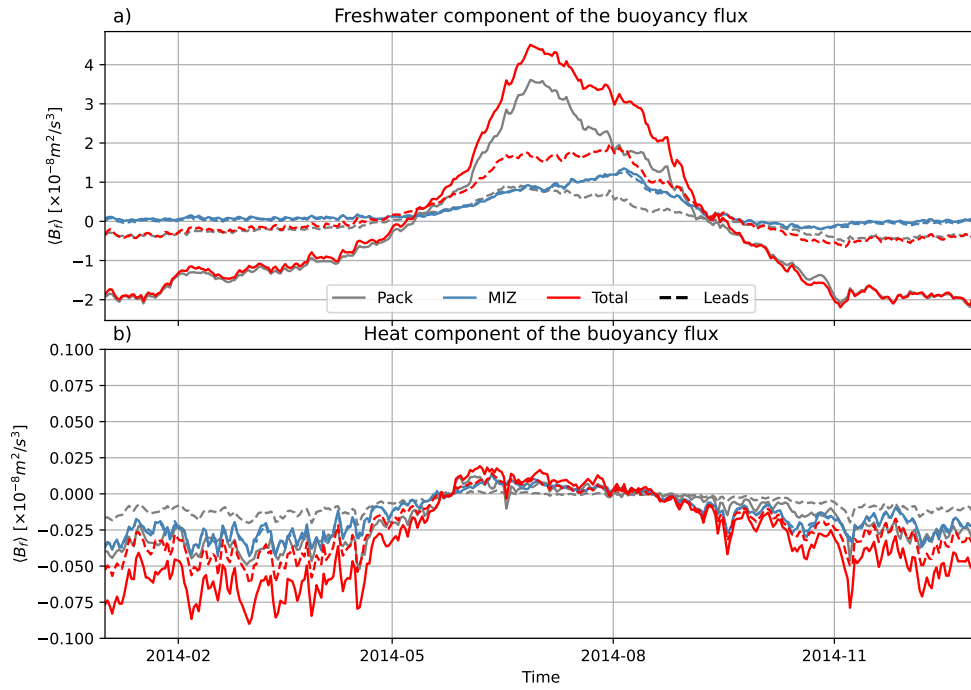


Fig. S2 Time-series during 2014 of a) the freshwater component of the buoyancy flux and b) the heat component of the buoyancy flux for the ice pack, the MIZ, and the leads in the ice pack and in the MIZ. The MIZ is defined between ice concentration of 15-80%, and the ice pack is defined where ice concentration > 80%.